

Impaired endothelial function following a meal rich in used cooking fat.

OBJECTIVES: The purpose of this study was to test the hypothesis that intake of used cooking fat is associated with impaired endothelial function. **BACKGROUND:** Diets containing high levels of lipid oxidation products may accelerate atherogenesis, but the effect on endothelial function is unknown. **METHODS:** Flow-mediated endothelium-dependent dilation and glyceryl trinitrate-induced endothelium-independent dilation of the brachial artery were investigated in 10 men. Subjects had arterial studies before and 4 h after three test meals: 1) a meal (fat 64.4 g) rich in cooking fat that had been used for deep frying in a fast food restaurant; 2) the same meal (fat 64.4 g) rich in unused cooking fat, and 3) a corresponding low fat meal (fat 18.4 g) without added fat. **RESULTS:** Endothelium-dependent dilation decreased between fasting and postprandial studies after the used fat meal ($5.9 \pm 2.3\%$ vs. $0.8 \pm 2.2\%$, $p = 0.0003$), but there was no significant change after the unused fat meal ($5.3 \pm 2.1\%$ vs. $6.0 \pm 2.5\%$) or low fat meal ($5.3 \pm 2.3\%$ vs. $5.4 \pm 3.3\%$). There was no significant difference in endothelium-independent dilation after any of the meals. Plasma free fatty acid concentration did not change significantly during any of the meals. The level of postprandial hypertriglyceridemia was not associated with change in endothelial function. **CONCLUSIONS:** Ingestion of a meal rich in fat previously used for deep frying in a commercial fast food restaurant resulted in impaired arterial endothelial function. These findings suggest that intake of degradation products of heated fat contribute to endothelial dysfunction.